

Natural Language Processing, HW2

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1. Method Description (10%)

- What model did you use and why?
- Describe your approach and explain the reasoning in detail.
- Show your exact prompt template(s).
- Detail any additional techniques (e.g., calibration, ensembling, fine-tuned, chain-of-thought).

Ans:

1.1 Model Selection

We used qwen3:8b as our judge model. The selection was based on the following reasons:

- Open-weight and within the 9B parameter constraint required by the assignment.
- Strong instruction-following capability, which is critical for structured output formats like 'VERDICT: A'.
- Supports think=False mode via Ollama, which disables internal chain-of-thought reasoning and significantly speeds up inference without sacrificing output quality for our structured prompting approach.

1.2.1 Approach Overview

Our approach evolved iteratively through multiple stages, progressively addressing each observed failure mode:

Stage	Method	Kaggle Score	Key Improvement
1	Zero-shot	0.460	Baseline — direct prediction with no examples
2	Zero-shot + Debiasing	0.467	Added forward/reverse voting to reduce position bias
3	CoT + Debiasing	0.517	Added step-by-step reasoning before verdict
4	Few-shot + CoT + Debiasing	0.543	Added labeled examples for all 4 verdict types

5	Few-shot + refined instructions	0.560	Tuned judgment step order (neither→A/B→tie)
6	Few-shot + balanced tie guidance	0.590	Balanced tie instruction: length ≠ quality

1.2.2 Detailed Reasoning

1. **Zero-shot:** The model was given both dialogs and asked to output A/B/tie/neither directly. This revealed severe position bias: A was predicted 79.2% of the time despite A and B having equal ground-truth frequency.
2. **Position Debiasing:** We introduced a three-pass voting scheme (forward pass, reverse pass with flip, and tiebreaker) to eliminate positional preference. This brought the A:B ratio from 4.8:1 down to nearly 1:1.
3. **Chain-of-Thought:** Adding step-by-step evaluation criteria (accuracy, helpfulness, clarity) before the verdict improved accuracy by forcing the model to reason before deciding, reducing impulsive positional choices.
4. **Few-shot Examples:** We added 9 labeled examples covering all four verdict types. Neither examples were sourced from actual train.json cases, categorized into 4 types: both wrong answers, both too short, both give the same wrong answer, and both refuse/dodge. This substantially improved neither detection.
5. **Judgment Step Order:** We changed the judgment order from [neither→tie→A/B] to [neither→A/B→tie], making tie the last resort. This prevented the model from defaulting to tie on borderline cases.
6. **Balanced Tie Guidance:** The final key insight was that tie was still being underused because the model equated 'longer = better'. Adding the instruction 'Use tie when both responses cover the same key points, even if one is slightly longer' corrected this length bias and brought the score to 0.590.

1.3 Final Prompt Template

The following is the complete prompt template used in our final submission:

```
You are a fair and impartial judge evaluating two AI assistant
responses.

Study these examples carefully before making your judgment:

=== NEITHER examples (both responses fail) ===

[Type 1 - Both give wrong answers]
User: Solve this crossword clue: "BLANK, I'm Adam" (palindrome, 5
letters)
Response A: I apologize for the mistake. The correct answer is
"SIS."
Response B: "EVEE" is 5 letters long and is a palindrome.
```

-> Both gave factually wrong answers. VERDICT: neither

[Type 2 - Both too short / unhelpful]

User: The following is a bash command. Please determine if it is safe.

Response A: NO

Response B: NO

-> Both responses are too vague and provide no useful explanation. VERDICT: neither

[Type 3 - Both give the same wrong answer]

User: Act as a professor grading proofs, did this proof pass?

Response A: Yes, this proof passed!

Response B: Yes, this proof passed the test

-> Both agree but are both incorrect. VERDICT: neither

[Type 4 - Both refuse or dodge the question]

User: Do not include numeric chars. What's the smallest non-negative integer?

Response A: The smallest non-negative integer is not defined.

Response B: The smallest non-negative integer is 0.

-> A is wrong; B violates the constraint. VERDICT: neither

=== NOT neither examples ===

[Counter-example 1 - This is A, not neither]

User: What causes rainbows?

Response A: Rainbows are caused by refraction and reflection of sunlight in water droplets.

Response B: Rainbows happen because of water in the air.

-> A is detailed; B is vague but not wrong. One is clearly better. VERDICT: A

[Counter-example 2 - This is tie, not neither]

User: What is the capital of Japan?

Response A: The capital of Japan is Tokyo.

Response B: Tokyo is Japan's capital city.

-> Both are correct and helpful. VERDICT: tie

=== A / B / tie examples ===

[VERDICT: A]

User: Explain what a for loop does in Python.

Response A: A for loop iterates over a sequence. Example: for i in [1,2,3]: print(i)

Response B: For loops are used in programming to repeat things.

-> A is more complete with a concrete example. VERDICT: A

[VERDICT: B]

User: What are the symptoms of dehydration?

Response A: Dehydration happens when you don't drink enough water.

Response B: Symptoms include thirst, dark urine, dizziness, fatigue, dry mouth.

-> B directly answers with specific symptoms. VERDICT: B

[VERDICT: tie]

User: What is photosynthesis?

Response A: Plants convert sunlight, water, and CO2 into glucose and oxygen.

Response B: Plants use light energy, CO2, and water to make sugar and oxygen.

-> Both are accurate and equally helpful. VERDICT: tie

Now evaluate the following:

<conversation_A>

{format_dialog(dialog_a)}

</conversation_A>

<conversation_B>

{format_dialog(dialog_b)}

</conversation_B>

Follow these steps:

Step 1 - Are BOTH responses wrong, unhelpful, or failing to answer?

-> neither

Step 2 - Is one response noticeably more accurate, complete, or helpful? -> A or B

Step 3 - Only if both are correct AND quality difference is truly negligible -> tie

Remember:

- Use 'neither' when BOTH responses fail, not just one
- Use 'A' or 'B' whenever one is noticeably better, even slightly
- Use 'tie' when both are correct and cover the same key points, even if one is slightly longer or more detailed
- A longer response is NOT automatically better
- Do NOT default to 'tie' when unsure -- pick the better one

After your analysis, output your final verdict on the last line:

VERDICT: [A or B or tie or neither]

Your analysis:

1.4 Additional Techniques

- **Position Debiasing (Three-pass Voting):** Each sample is evaluated three times. The first pass presents dialog_1 as A and dialog_2 as B. The second pass swaps the order, and the result is flipped back to the original reference frame. If both passes agree, the result is adopted directly. If they disagree, a third tiebreaker pass runs with temperature=0.3 to introduce slight randomness and break the deadlock.

Pass	Order	Temperature	Purpose
1 (Forward)	dialog_1=A, dialog_2=B	0	Primary prediction
2 (Reverse)	dialog_1=B, dialog_2=A	0	Detect position bias
3 (Tiebreaker)	dialog_1=A, dialog_2=B	0.3	Resolve disagreements

- **Chain-of-Thought Prompting:** The prompt instructs the model to analyze both responses step by step before giving the final verdict. The model writes out its reasoning before outputting 'VERDICT: X', which reduces impulsive positional choices and improves accuracy on borderline cases.
- **Robust Verdict Parsing:** Since few-shot prompts contain multiple VERDICT labels in the examples, the parser extracts the last occurrence of 'VERDICT: X' in the model output to avoid being misled by example labels. A multi-level fallback (last line, second-half scan) ensures robustness when the model deviates from the expected format.

2. Comparative Error Analysis (10%)

Implement at least 2 different judging approaches (e.g., zero-shot vs. chain-of-thought, different prompt designs, or different models) and compare their failure modes on the training set.

- Report each method's accuracy on a held-out split of train.json.
- For each method, identify its most common error pattern (e.g., errors concentrated on tie cases, long responses, ambiguous instructions) and provide one concrete example.

- Explain why different methods fail differently — what does each method's error pattern reveal about its underlying weakness?

Ans:

2.1 Methods

- Method 1: Zero-shot Prompting — the model is given the two responses and directly asked to output A, B, tie, or neither without any examples or reasoning steps.
- Method 2: Few-shot Chain-of-Thought (CoT) with Position Debiasing — the model is given 9 labeled examples covering all verdict types, asked to reason step by step, and predictions are debiased using a forward/reverse/tiebreaker voting scheme.

2.2 Error Pattern

Method	Val Accuracy	Submission Score
Zero-shot	~46%	0.460
Few-shot + CoT + Debiasing	~57%	0.543

- Method 1: Zero-shot

Most common error pattern: Position Bias

The zero-shot model systematically favored the first response (Response A), regardless of actual quality. In our submission, A was predicted 79.2% of the time despite the true distribution being nearly 50/50 between A and B.

Example (id=11, gold=B):

User: How far is the earth from the moon?

Response A: The distance is approximately 384,442 miles (657,341 km). This is the average distance between the two planets...

Response B: The average distance is about 384,400 km (238,855 miles). However, the Moon's orbit is not a perfect circle...

Gold verdict: B | Zero-shot prediction: A

Response B is more accurate, but the zero-shot model chose A simply because it appeared first in the prompt. This reflects a well-known position bias in LLMs trained on human preference data, where annotators tend to favor the first option they read.

- Method 2: Few-shot CoT + Debiasing

Most common error pattern: Underestimating 'neither' cases

Despite significant improvement over zero-shot, the few-shot model still struggles to correctly identify cases where both responses are bad. In our submission, neither was predicted only 5.9% of the time, while the true proportion is ~18.3%.

Example (id=34, gold=neither):

User: Please sort alphabetically: azertyuiop

Response A: azertyuiop

Response B: A Z E R T Y U I O P

Gold verdict: neither | Few-shot prediction: B

Neither response correctly sorts the characters alphabetically. Response A returns the original unsorted string. Response B capitalizes and spaces the letters but does not sort them. The correct verdict is neither, but the model incorrectly chose B, likely because B appears more "structured" and "effort-ful" than A.

2.3.1 Why Do the Methods Fail Differently?

	Zero-shot	Few-shot + CoT + Debiasing
Primary failure	Position bias (prefers A)	Underestimates 'neither'
Root cause	No reasoning process; relies on surface-level position heuristic	Model trained to be helpful; reluctant to say both responses are bad
Error concentration	Errors on gold=B cases (selected A instead)	Errors on gold=neither cases (selected A or B instead)
tie prediction	Very rare (0.6%)	Moderate (9.7%), still below true 10.9%

2.3.2 Cause

The zero-shot model's failure reveals a positional heuristic: without explicit reasoning, the model defaults to the first option it sees; the few-shot CoT model's failure reveals a helpfulness bias: having been trained to assist users, the model is reluctant to conclude that both responses are worthless, and instead picks the "less bad" one.

3. Position Bias Analysis (10%)

Position bias is a well-known issue in pairwise LLM evaluation — models may systematically prefer the response in a particular position (first or second), regardless of quality.

- Design an experiment using the training data to measure whether your model exhibits position bias.
- Report quantitative results (e.g., how often does your model pick A vs B?).

- If you implemented any debiasing strategies, describe them and evaluate their effectiveness.

Ans:

3.1 Experiment Design

Steps:

1. Present dialog_1 as Response A and dialog_2 as Response B. Record the model's prediction.(Forward)
2. Swap the order — present dialog_2 as Response A and dialog_1 as Response B. Record the model's prediction.(Reverse)
3. Invert the reverse prediction (A→B, B→A, tie→tie, neither→neither) so both passes use the same reference frame.
4. Compare forward and flipped-reverse predictions. Inconsistencies indicate position bias.

We ran this experiment on 200 validation samples using qwen3:8b.

3.2 Quantitative Results

Result 1 — Zero-shot model prediction distribution (1000 test samples):

Verdict	Predicted Count	Predicted %	True Distribution %
A	792	79.2%	35.5%
B	165	16.5%	35.2%
tie	6	0.6%	10.9%
neither	37	3.7%	18.3%

The zero-shot model predicted A 79.2% of the time despite A and B being nearly equally distributed in the ground truth (~35% each). This is a clear sign of systematic position bias toward the first response.

Result 2 — Forward vs. Reverse consistency (200 val samples):

Metric	Value
Consistency rate (forward == flipped reverse)	81.5%
Inconsistency rate (position bias detected)	18.5%
Cases sent to tiebreaker	37 / 200

81.5% of predictions were consistent across both orderings, meaning the model gave the same verdict regardless of which response appeared first. The remaining 18.5% were inconsistent, confirming that position bias still affects the model in roughly 1 in 5 cases even after applying few-shot CoT.

Result 3 — A vs. B prediction ratio comparison:

Method	A Predicted	B Predicted	A:B Ratio
Zero-shot (no debiasing)	792	165	4.8 : 1
Few-shot + Debiasing	410	434	0.94 : 1
True distribution	~355	~352	~1 : 1

After applying position debiasing, the A:B ratio dropped from 4.8:1 to nearly 1:1, closely matching the true distribution. This confirms that the debiasing strategy effectively eliminates the systematic preference for the first position.

3.3.1 Debiasing Strategy and Effectiveness

We implemented a three-pass voting scheme to debias predictions:

```

Pass 1 (Forward):  dialog_1=A, dialog_2=B → predict verdict_1
Pass 2 (Reverse):  dialog_1=B, dialog_2=A → predict verdict_2_raw
                  flip verdict_2_raw      → verdict_2

If verdict_1 == verdict_2:
    Final = verdict_1 (consistent, adopt directly)
Else:
    Pass 3 (Tiebreaker, temperature=0.3): → verdict_3
    If verdict_3 == verdict_1: Final = verdict_1
    If verdict_3 == verdict_2: Final = verdict_2
    Else:                      Final = 'tie' (cannot resolve)

```

The key insight is that if the model truly prefers response A because of quality (not position), swapping the order should still yield a preference for the same response. Only if both forward and flipped-reverse predictions agree do we trust the verdict without further verification.

3.3.2 Effectiveness summary:

Strategy	A:B Ratio	Val Accuracy	Kaggle Score
No debiasing (zero-shot)	4.8:1	~46%	0.460
Debiasing + Tiebreaker	~1:1	~50%	0.467
Few-shot + CoT + Debiasing	0.94:1	~57%	0.543

Position debiasing alone improved the A:B ratio dramatically but only modestly improved accuracy (+0.7%), because the main remaining bottleneck shifted to the model's difficulty distinguishing neither and tie cases. The largest accuracy gain came from combining debiasing with few-shot CoT prompting, which improved the model's overall reasoning quality.